

AhamX Bodhi

Where I Becomes Infinite

Audio Processing Fundamentals

Module 4.2: Audio and Speech Models

Prof. Sashikumaar Ganesan | IISc Bengaluru | 2026



Session Overview

What You Will Learn

- Digital audio: sampling, quantization, PCM
- Time-domain vs. frequency-domain representation
- Fourier Transform and Short-Time FT
- Mel scale and perceptual audio features
- Engineering audio pipelines for AI systems

Concept at a Glance

- **Duration:** 9 minutes
- **Bloom's Level:** B2 – Understand
- **Difficulty:** L2 – Intermediate
- **Module:** 4.2 Audio and Speech Models
- **Prerequisite:** Module 4.1 concepts





From Sound Waves to Digital Signals

Digital Audio: Sampling and Quantization

- **Sound:** Continuous pressure wave varying over time. Frequency (Hz) → perceived pitch; amplitude → perceived loudness.
- **Sampling:** Measure the waveform amplitude at regular intervals. *Sample rate* (Hz) = measurements per second. Standard: 16kHz (speech), 44.1kHz (music), 48kHz (broadcast).
- **Nyquist theorem:** Sample rate must be $\geq 2 \times$ max frequency to reconstruct the signal without aliasing. Human speech: 300Hz–8kHz → 16kHz sufficient.
- **Quantization:** Represent amplitude as integer. 16-bit PCM = 65,536 amplitude levels. CD quality: 44.1kHz, 16-bit, stereo = 1.4 Mbps raw.
- **AI standard:** Most speech models (Whisper, wav2vec 2.0) expect mono, 16kHz, 16-bit PCM audio.

Audio preprocessing is a critical production step – resampling a 44.1kHz stereo recording to 16kHz mono is required before feeding to most speech AI models.



Time Domain vs. Frequency Domain

Two Views of the Same Signal

- **Time domain:** Raw waveform – amplitude at each time step. Easy to compute, hard to extract meaningful features. “When” something happens.
- **Frequency domain:** How much energy at each frequency. Computed via Fast Fourier Transform (FFT). “What frequencies” are present.
- **Why frequency domain?** Speech content (phonemes) is encoded in frequency structure, not raw amplitude. Neural networks learn more efficiently from frequency features.
- **Limitation of FFT:** Global transform – loses temporal information. Cannot tell *when* a frequency occurred in the signal.
- **Solution:** Short-Time Fourier Transform (STFT) – apply FFT on overlapping windows (“frames”) to get a time-frequency representation.

The STFT is the foundation of virtually all audio features used in deep learning – spectrogram, mel-spectrogram, MFCC all derive from it.



Short-Time Fourier Transform and Spectrogram

STFT: Time-Frequency Representation

- **STFT process:** Slide a window (e.g., 25ms, Hann window) over the waveform with a hop size (e.g., 10ms); apply FFT to each frame
- **Output:** Complex matrix of shape [freq_bins, time_frames]. For 16kHz audio with 512-point FFT: 257 frequency bins.
- **Spectrogram:** Magnitude of STFT (discard phase): $|\text{STFT}(t, f)|^2$. Visualized as image: x-axis = time, y-axis = frequency, color = energy.
- **Key parameters:** Window size (FFT-N), hop length, window function (Hann, Hamming). Larger window $\rightarrow$ better frequency resolution but worse time resolution (Heisenberg trade-off).
- **Log spectrogram:** Apply $\log(1 + |S|^2)$ to compress dynamic range – human auditory system is logarithmic. Significantly improves model performance.

The spectrogram transforms a 1D audio waveform into a 2D image – this is why CNN and ViT architectures designed for images can be applied to audio.





The Mel Scale: Perceptually Motivated Frequency Axis

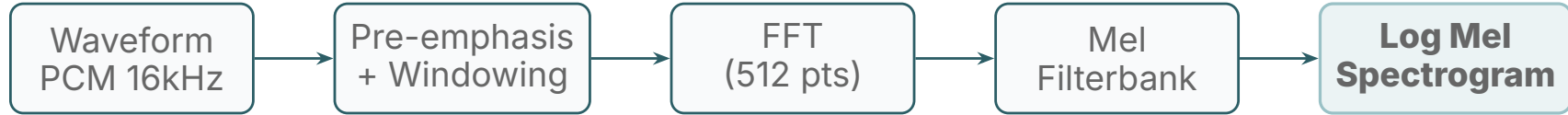
Why Linear Frequency Is Wrong for Speech

- **Human hearing:** Humans perceive pitch differences on a logarithmic scale – doubling from 100Hz to 200Hz sounds like the same interval as 1000Hz to 2000Hz
- **Mel scale:** A perceptual scale of pitches where equal intervals sound equally different. $m = 2595 \cdot \log_{10}(1 + f/700)$
- **Mel filterbank:** Apply B triangular filters (e.g., 80 or 128) spaced on the Mel scale to the power spectrogram. Each filter outputs the energy in that perceptual frequency band.
- **Mel-spectrogram:** Output of mel filterbank – shape `[mel_bins, time_frames]`. E.g., 80 mel bins $\times$ 3000 frames for a 30-second clip.
- **Why it works:** Mel filterbanks emphasize speech-critical frequencies (300–3000Hz) and compress less informative high frequencies – aligned with human auditory perception.

Whisper (OpenAI) uses 80-bin log mel-spectrogram as input. Most modern ASR and TTS models use mel-spectrograms as the intermediate audio representation.



Audio Feature Extraction Pipeline



Standard parameters for ASR (Whisper-compatible): sample rate 16kHz, window 25ms, hop 10ms, FFT size 512, mel bins 80, normalize to $[-1, 1]$.



MFCCs: Mel-Frequency Cepstral Coefficients

MFCCs: Classic Speech Feature Engineering

- **Process:** Log mel-spectrogram $\rightarrow$ Discrete Cosine Transform (DCT) $\rightarrow$ take first C coefficients (typically 13 or 40)
- **What DCT does:** Decorrelates mel filterbank channels; lower coefficients capture spectral shape (phoneme identity), higher coefficients capture fine spectral details
- **Delta features:** First and second temporal derivatives (Δ , Δ^2) appended to capture dynamics – standard MFCC feature is $39 = 13 \times 3$ dimensions
- **Historical importance:** Standard feature for HMM-GMM ASR systems (1980s–2010s). Still used in speaker verification and emotion recognition.

MFCC vs. Mel-Spectrogram for Deep Learning

- Modern neural ASR (Whisper, wav2vec) uses mel-spectrograms directly – DCT compression is no longer needed when the model can learn its own features
- MFCCs remain useful for lightweight applications (on-device keyword detection, speaker ID)





GenAI Application: Audio Preprocessing at Scale

Building Production Audio Ingestion Pipelines

- **Input diversity:** Real-world audio arrives at 8kHz (phone), 16kHz (microphone), 44.1kHz (music), 48kHz (video) – all must be normalized before AI processing
- **Tools:** librosa (Python, research), torchaudio (GPU-accelerated), FFmpeg (format conversion), sox (command-line preprocessing)
- **Chunking strategy:** Most ASR models have max context (Whisper: 30 seconds). Long audio must be split with silence detection (VAD) to avoid splitting mid-sentence
- **Voice Activity Detection (VAD):** Silero-VAD (lightweight, runs on CPU) detects speech segments – discard silence before ASR to reduce cost by 30–60%
- **Noise robustness:** Apply spectral subtraction or denoising (DeepFilterNet, DTLN) before ASR when audio quality is poor (call centers, field recordings)

Audio preprocessing quality directly determines ASR accuracy – a well-tuned preprocessing pipeline can improve WER by 10–20% with no model changes.



GenAI Application: Audio Fingerprinting and Recognition

Shazam-Style Audio Identification with Spectrograms

- **Shazam algorithm:** Compute spectrogram, find local peak constellations (time-frequency anchor points), hash pairs of peaks into a fingerprint. Match against 10M+ song database in milliseconds.
- **Neural approach:** Encode audio segment with CNN or Transformer encoder; compare against indexed embeddings. Neural Audio Fingerprint (Chang et al., 2021) achieves 99.9%+ accuracy on 10-second clips even with noise and distortion
- **Industrial scale:** YouTube Content ID uses audio fingerprinting to match >800 years of video uploaded per day against a 100M+ reference database
- **AI engineer's role:** Build the embedding index (FAISS), design the audio encoder, optimize ANN search for sub-100ms latency at scale

Key Insight

Spectrogram-based fingerprinting is robust to re-encoding, tempo changes, and low SNR – properties that make it practical for real-world audio matching.



GenAI Application: Speaker Diarization

"Who Spoke When?" in Meeting Transcription

- **Task:** Segment an audio recording by speaker identity – essential for meeting notes, call center analysis, legal transcription
- **Pipeline:** VAD → MFCC/mel feature extraction → speaker embedding (x-vector, ECAPA-TDNN) → agglomerative clustering → speaker labels
- **State of the art:** pyannote-audio 3.0 + WhisperX achieves $<5\%$ DER (Diarization Error Rate) on AMI meeting corpus with speaker identification
- **Production system:** AWS Transcribe, Google Speech-to-Text, and AssemblyAI all offer integrated diarization APIs built on these fundamentals
- **Audio engineer's challenge:** Overlapping speech (two speakers simultaneously) remains difficult – current best systems degrade significantly at $>20\%$ overlap

Understanding STFT and mel features is prerequisite knowledge for debugging diarization systems – most errors trace back to feature quality or VAD failures.



Key Takeaways

What You Should Remember

- Digital audio is a sequence of amplitude samples (PCM); standard AI speech models expect 16kHz mono 16-bit input
- STFT converts 1D waveforms to 2D time-frequency representations; mel filterbanks compress the frequency axis to match human perception
- Log mel-spectrogram (80 bins, 10ms hop) is the de facto input representation for modern ASR (Whisper) and TTS (Tacotron, VITS) models
- MFCCs remain useful for lightweight speaker ID and keyword detection; mel-spectrograms preferred for deep learning models
- Production pipeline: VAD → resampling → denoising → feature extraction → chunking → model inference

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